SECTION

REGRESSION TECHNIQUES

5

CHAPTER

21 CALCULATING THE SOLUTION FOR REGRESSION TECHNIQUES: PART 1—MULTIVARIATE REGRESSION MADE SIMPLE

For the next several chapters we will illustrate the straightforward calculations used for MLR, PCR, PLS, and SVD. In all cases we will use the same notation and perform all mathematical operations using MATLAB software [1,2]. We have already discussed and shown many of the manual methods for calculating the matrix algebra in references [3–6].

Let us begin by identifying a simple data matrix denoted by $\boldsymbol{A}$. $\boldsymbol{A}$ is used to represent a set of absorbances for three samples as a rows × columns matrix. For our example each row represents a different sample spectrum, and each column a different data channel, absorbance, or frequency. We arbitrarily designate $\boldsymbol{A}$ for our example as:

$$\boldsymbol{A} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix} = \boldsymbol{A}_{I\times K} = \begin{bmatrix} 1 & 7 \\ 4 & 10 \\ 6 & 14 \end{bmatrix} \tag{21-1}$$

Thus, the integers 1 and 7 represent the instrument signal for two data channels (frequencies 1 and 2) for sample Spectrum #1, 4 and 10 represent the same data channel signals (e.g., frequencies 1 and 2) for sample Spectrum #2, and so on.

If we arbitrarily set our concentration $\boldsymbol{c}$ vector representing a single component to be a single column of numbers as:

$$\boldsymbol{c}_{r\times c} = \begin{bmatrix} c_{11} \\ c_{21} \\ c_{31} \end{bmatrix} = \boldsymbol{c}_{I\times 1} = \begin{bmatrix} 4 \\ 8 \\ 11 \end{bmatrix} \tag{21-2}$$

We now have the data necessary to calculate the matrix of regression coefficients $\boldsymbol{b}$ which is given by

$$\boldsymbol{b} = \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} = (A'A)^{-1}A'c = A^{+}c = \hat{p} \tag{21-3}$$

This $\boldsymbol{b}$ (also known as $\hat{\boldsymbol{p}}$ = the *prediction vector*) is often referred to as the *regression vector* or set of *regression coefficients*. Note that $(\boldsymbol{A}'\boldsymbol{A})^{-1}\boldsymbol{A}'$ is referred to as the *pseudoinverse* of $\boldsymbol{A}$ designated as $\boldsymbol{A}^{+}$. Note that there is one regression coefficient for each frequency (or data channel).

The matrix of predicted values is easily obtained as matrix $\boldsymbol{A}$ (the data matrix) × vector $\boldsymbol{b}$ (the regression coefficients) = vector $\boldsymbol{c}$ (the predicted values). This is shown in matrix notation as:

$$\boldsymbol{A} \times \boldsymbol{b} = \boldsymbol{c} \tag{21-4}$$

Now using the MATLAB command line software, we can easily demonstrate this solution (for the multivariate problem we have identified) using a series of simple matrix operations as shown in Table 21-1.

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Table 21-1 Matrix Operations in MATLAB to Compute Eqs. (21-1)–(21-4)

Command Line	Comments
$\gg A = [1\ 7; 4\ 10; 6\ 14]$	Enter the $\boldsymbol{A}$ matrix
$\gg A = \begin{bmatrix} 1 & 7 \\ 4 & 10 \\ 6 & 14 \end{bmatrix}$	Display the $\boldsymbol{A}$ matrix
$\gg c = [4; 8; 11]$	Enter the concentration vector $\boldsymbol{c}$
$c = \begin{bmatrix} 4 \\ 8 \\ 11 \end{bmatrix}$	Display the concentration vector $\boldsymbol{c}$
$\gg b = \text{inv}(A'*A)*A'*c$	Calculate the regression vector [Note: The inverse applies only to $(A'*A)$]
$b = \begin{bmatrix} 0.7722 \\ 0.4662 \end{bmatrix}$	Display the regression vector $\boldsymbol{b}$
$\gg A*b$ $ans = \begin{bmatrix} 4.0356 \\ 7.7509 \\ 11.1601 \end{bmatrix}$	Predict the concentrations [Note: A residual (or difference) exists between the predicted and the actual concentrations 4, 8, and 11]

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CHAPTER

22 CALCULATING THE SOLUTION FOR REGRESSION TECHNIQUES: PART 2—PRINCIPAL COMPONENT(S) REGRESSION MADE SIMPLE

For the next several chapters in this book, we will illustrate the straightforward calculations used for multivariate regression. In each case, we continue to perform all mathematical operations using MATLAB software [1,2]. We have already discussed and shown the manual methods for calculating most of the matrix algebra used here in Refs. [3–6]. You may wish to program these operations yourselves or use other software to routinely make these calculations.

As in Chapter 21, we begin by identifying a simple data matrix denoted by $\boldsymbol{A}$. $\boldsymbol{A}$ is used to represent a set of absorbances for three samples as a rows × columns matrix. For our example, each row represents a different sample spectrum, and each column a different data channel, absorbance, or frequency. We arbitrarily designate $\boldsymbol{A}$ for our example as:

$$\boldsymbol{A} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix} = \boldsymbol{A}_{I\times K} = \begin{bmatrix} 1 & 7 \\ 4 & 10 \\ 6 & 14 \end{bmatrix} \tag{22-1}$$

Thus, 1 and 7 represent the instrument signal for two data channels (frequencies 1 and 2) for sample spectrum #1; 4 and 10 represent the same data channel signals (e.g., frequencies 1 and 2) for sample spectrum #2; and so on.

We now have the data necessary to calculate the *singular value decomposition* (SVD) for matrix $\boldsymbol{A}$. The operation performed in SVD is sometimes referred to as *eigenanalysis*, PCA, or *factor analysis*. If we perform SVD on the $\boldsymbol{A}$ matrix, the result is three matrices, termed the "left singular values" (LSV) matrix or the "$\boldsymbol{U}$ matrix"; the "singular values matrix" (SVM) or the "$\boldsymbol{S}$ matrix"; and the "right singular values matrix" (RSV) or the "$\boldsymbol{V}$ matrix."

We now have enough information to find our Scores matrix and Loadings matrix. First of all the Loadings matrix is simply the RSV matrix or the $\boldsymbol{V}$ matrix; this matrix is referred to as the $\boldsymbol{P}$ matrix in PCA terminology. The Scores matrix is calculated as:

$$\text{The data matrix } \boldsymbol{A} \times \text{the Loadings matrix } \boldsymbol{V} = \text{Scores matrix } \boldsymbol{T} \tag{22-2}$$

Note: The Scores matrix is referred to as the $\boldsymbol{T}$ matrix in PCA terminology. Let us look at what we have completed so far by showing the SVD calculations in MATLAB as illustrated in Table 22-1.

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Table 22-2 Matrix Operations in MATLAB to Compute Eqs. (22-4)–(22-6)

Command Line	Comments
$\gg \boldsymbol{A}\text{est} = \boldsymbol{T}^*\boldsymbol{V}'$	Estimate the $\boldsymbol{A}$ data matrix
$\boldsymbol{A}\text{est} = \begin{matrix} 1.0000 & 7.0000 \\ 4.0000 & 10.0000 \\ 6.0000 & 14.0000 \end{matrix}$	Display the estimate for $\boldsymbol{A}$
$\gg \boldsymbol{b} = \boldsymbol{V}(:,1{:}2)^*\text{inv}.(\boldsymbol{S}(1{:}2,1{:}2))^*\boldsymbol{U}(:,1{:}2)'^*\boldsymbol{c}$;	Calculate the regression vector [Note: The inverse operation refers only to the singular values matrix $\boldsymbol{S}$. The calculation to determine $\boldsymbol{b}$ can only be performed using two columns in each of the $\boldsymbol{V}$, $\boldsymbol{S}$, and $\boldsymbol{U}$ matrices; this number is equivalent to the number of latent variables (or principal components) used]
$\boldsymbol{b} = \begin{matrix} 0.7722 \\ 0.4662 \end{matrix}$	Display the regression vector
$\gg \boldsymbol{c}\text{est} = (\boldsymbol{T}^*\boldsymbol{V}')^*\boldsymbol{b}$	Predict the concentrations [Note: This computation is equivalent to $(\boldsymbol{A}\text{est} \times \boldsymbol{b})$]
$\boldsymbol{c}\text{est} = \begin{matrix} 4.0356 \\ 7.7509 \\ 11.1601 \end{matrix}$	Display the concentration vector [Note: For this example of PCR a residual (or difference) exists between the predicted and the actual concentrations 4, 8, and 11]

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CHAPTER

23 CALCULATING THE SOLUTION FOR REGRESSION TECHNIQUES: PART 3—PARTIAL LEAST SQUARES REGRESSION MADE SIMPLE

For the past three chapters we have described the most basic calculations for MLR, PCR, and PLS. Our intent is to show basic computations for these regression methods while avoiding unnecessary complexity which could confuse rather than instruct. There are of course a number of difficulties in taking this simplistic approach, namely the assumptions made for our simple cases do not always hold, and poorly behaved matrices are the rule rather than the exception. We have not yet discussed the concepts of rank, collinearity, scaling, or data conditioning. Issues of graphical representation and details of computational methods and assessing model performance are forthcoming. We ask that you abide with us over the next several chapters as we intend to delve much more deeply into the details and problems associated with regression methods.

For this chapter we will illustrate the straightforward calculations used for PLSR utilizing SVD. For PLS, a special case of SVD is used. You will notice that the PLS form of SVD includes the use of the concentration vector $\boldsymbol{c}$ as well as the data matrix $\boldsymbol{A}$. The reader will note that the scores and loadings are determined using the concentration values for PLS SVD whereas only the data matrix $\boldsymbol{A}$ is used to perform SVD for PCA. The SVD and PLS SVD will be the subject of several future chapters so we will only introduce its use here and not its derivation. All mathematical operations are completed using MATLAB software [1,2]. As previously discussed, the manual methods for calculating the matrix algebra used within these chapters on the subject are found in Refs. [3–7]. You may wish to program these operations yourselves or use other software to routinely make the calculations.

As in our last installment, we begin by identifying a simple data matrix denoted by $\boldsymbol{A}$. $\boldsymbol{A}$ is used to represent a set of absorbances for three samples and three data channels, as a rows × columns matrix. For our example each row represents a different sample spectrum, and each column a different data channel, absorbance, or frequency. We arbitrarily designate $\boldsymbol{A}$ for our example as:

$$\boldsymbol{A}_{r\times c}=\begin{bmatrix}A_{11} & A_{12} & A_{13}\\ A_{21} & A_{22} & A_{23}\\ A_{31} & A_{32} & A_{33}\end{bmatrix}=\boldsymbol{A}_{I\times K}=\begin{bmatrix}1 & 7 & 9\\ 4 & 10 & 12\\ 6 & 14 & 16\end{bmatrix} \tag{23-1}$$

Thus, 1, 7, and 9 represent the instrument signal for three data channels (frequencies 1, 2, and 3) for sample spectrum #1; 4, 10, and 12 represent the same data channel signals (e.g., frequencies 1, 2, and 3) for sample spectrum #2; and so on.

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CHAPTER

24 CALCULATING THE SOLUTION FOR REGRESSION TECHNIQUES: PART 4—SINGULAR VALUE DECOMPOSITION

In Chapters 21–23 and in this chapter, we have described the most basic calculations for MLR, PCR, and PLS. To reiterate, our intention is to demonstrate these basic computations for each mathematical method presently and then to delve into greater detail as the chapters progress; consider these articles' linear algebra bytes. For this chapter, we will illustrate the basic calculation and mathematical relationships of different matrices for the calculations of SVD.

You will note from previous chapters that SVD is used for modern computations of PCR and PLSR, although slightly different forms of SVD are used for each set of computations. Recall for PCR we simply used SVD and for PLS a special case of SVD that we called PLS SVD was used. You will also recall that the PLS form of SVD includes the use of the concentration vector $\boldsymbol{c}$ *as well as* the data matrix $\boldsymbol{A}$. The reader will note that the Scores ($\boldsymbol{T}$) and Loadings ($\boldsymbol{V}$) are determined using the concentration values for PLS SVD whereas only the data matrix $\boldsymbol{A}$ is used to perform SVD for PCA. All mathematical operations used for this chapter are completed using MATLAB software for Windows [1]. As previously discussed the manual methods for calculating the matrix algebra used within these chapters are found in Refs. [2–5]. You may wish to program these operations yourselves or use other software to routinely make the calculations.

As in previous installments we begin by identifying a simple data matrix denoted by $\boldsymbol{A}$. $\boldsymbol{A}$ is used to represent a set of absorbances for three samples and three data channels, as a rows × columns matrix. For our example each row represents a different sample spectrum and each column a different data channel, absorbance, or frequency. We arbitrarily designate $\boldsymbol{A}$ for our example as:

$$\boldsymbol{A}_{r\times c} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} = \boldsymbol{A}_{I\times K} = \begin{bmatrix} 1 & 7 & 9 \\ 4 & 10 & 12 \\ 6 & 14 & 16 \end{bmatrix} \tag{24-1}$$

Thus, 1, 7, and 9 represent the instrument signal for three data channels (frequencies 1, 2, and 3) for sample spectrum #1; 4, 10, and 12 represent the same data channel signals (e.g., frequencies 1, 2, and 3) for sample spectrum #2; and so on.

Given any data matrix $\boldsymbol{A}$ of arbitrary size (as rows × columns) the matrix $\boldsymbol{A}$ can be written or defined using the computation of SVD [6–8] as:

$$\boldsymbol{A} = \boldsymbol{USV}' = \boldsymbol{U} \times \boldsymbol{S} \times \boldsymbol{V}' \tag{24-2}$$

where $\boldsymbol{U}$ is the LSV matrix, $\boldsymbol{V}$ is the Loadings matrix, and $\boldsymbol{S}$ is the diagonal matrix containing information on the variance described by each PC (as the $\boldsymbol{S}$ matrix columns). It is important to note when

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Table 24-1 Simple SVD Performed on Matrix *A* Using MATLAB; Other Matrix Relationships Are Also Shown (See Eqs. 24-1–24-4)—cont'd

Command Line	Comments
≫***U***S* 11.3024 1.8038 −0.0370 *ans* = 16.1231 −0.1243 0.1704 22.0748 −0.8328 −0.1055	As mentioned in the text of the article, the scores matrix ***T*** can also be calculated as ***U*** × ***S***
≫***T***V*′ 1.0000 7.0000 9.0000 *ans* = 4.0000 10.0000 12.0000 6.0000 14.0000 16.0000	As we have stated, the original data matrix ***A*** can be estimated as the scores matrix (***T***) × the transpose of the loadings matrix (***V***′) as shown
≫***A***V***V*′ 1.0000 7.0000 9.0000 *ans* = 4.0000 10.0000 12.0000 6.0000 14.0000 16.0000	Just another way to estimate the original data matrix ***A***. In this case, ***V*** times the transpose of ***V*** (itself) is a diagonal matrix with a value of ones along the diagonal, such as shown below. Note: This matrix of ones along the diagonal is called an identity matrix or (***I***). 1.0000 0.0000 0.0000 0.0000 1.0000 0.0000 0.0000 0.0000 1.0000

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CHAPTER

25 INTERLUDE: LOOKING BEHIND AND AHEAD☆

We depart from discussion of our usual topics in this chapter. Over the years since beginning writing on this topic, there has been a spate of telephone calls where the callers, after introducing themselves, said something that could generically be rendered as: "By chance I came across a copy of one of your articles, and am interested in reading more about this subject. Are there any more articles like this, and what are they, and how can I get them?"

After discussing this between ourselves, we decided that we have reached a point where it is worthwhile to present our readers with a complete set of the chemometrics writings published to date. Those of you who have been reading our work for a long time will recall that the column series "Chemometrics in Spectroscopy" is a continuation of our previous column series, "Statistics in Spectroscopy." The column "Statistics in Spectroscopy" was published from 1986 to 1992, with some preliminary articles in 1985. The columns from this earlier series, "Statistics in Spectroscopy," as has been mentioned several times in the previous chapter, have been collected and published in their entirety as a book (with minor editorial changes appropriate to the change in format from a series of columns to a book, just as we are doing in this book) of the same name, now in its second edition [1]. It is, however, still available from on-line booksellers such as amazon.com and barnesandnobel.com. Without being experts in the publishing industry, our understanding is that publishing has changed over the years. The publication model for small-circulation books (what's a "small-circulation" book? Anything fewer than a good fraction of millions of copies, as far as we know) such as "Statistics in Spectroscopy" and "Chemometrics in Spectroscopy" is printed using a "print-on-demand" publishing model. Thus it need never go "out of print" as long as the digital master copy exists.

The original Spectroscopy columns themselves are no longer available. The publisher does not retain back issues of the print version for more than a fairly brief time. They do maintain digital copies and an online archive on their website (www.spectroscopyonline.com), but that contains only issues published for the past 15 years. So, except for those copies retained by individuals (such as the present authors, who have a complete set of issues of Spectroscopy), the original magazines are no longer available. Indeed, that was one of the key reasons we decided to preserve the contents of those columns in a less ephemeral format and developed them as books as well as in the original magazines.

So much for the past; what about the discussion?

The last few chapters have been presenting the "nuts and bolts" of some of the more common chemometric techniques for performing quantitative chemometric/spectroscopic calibration, even getting down to the level of a "cookbook" of actual code (written for the MATLAB Matrix Algebra

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multivariate analysis software). Subsequent chapters will deal first with extending and completing a discussion on the various chemometric techniques in current use. In the future we will go "under the hood" of the algorithms, emphasize the underlying mathematical and theoretical framework that these methods rest upon.

One topic which then followed in the series of columns, but was already included in previous chapters in this book, was description of the so-called "statistical design of experiments" methodologies, emphasizing those techniques that tend to be obscure but are more useful than they are dealt with in mainstream Chemometric discussions.

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CHAPTER

26 A SIMPLE QUESTION☆

In a recent article [1], Steve Brown and Barry Lavine state: "Chemometrics is not a subfield of Statistics. Although statistical methods are employed in Chemometrics, they are not the primary vehicles for data analysis." Parenthetically, we recommend this article as a very nice nonmathematical introduction for the average chemist as to what chemometrics is and how it can be used. As far as the quote is concerned, we have to both agree and disagree. On the one hand, we have to recognize the de facto truth that many users of chemometric techniques are not aware of the statistical backgrounds of the techniques, and indeed, we sometimes suspect that even the developers of those techniques may also not be aware of or, at least, give the statistical considerations their proper weight. Having said that, we will issue some disclaimers a little further on because there are some legitimate and justifiable reasons for the existence of this situation. However, ignoring the existence of this situation means that nobody is paying the attention that would eventually lead to the condition being corrected, which would result in a better theoretical understanding of the techniques themselves, with a concomitant improvement in their reliability and definition of their range of applicability.

This leads us to the other hand, which, it should be obvious, is that we feel that chemometrics *should* be considered a subfield of statistics, for the reasons given earlier. In addition, as we will see later on in this book, chemometrics are based on, or even taken from, methods developed by statisticians hundreds of years ago. Questions currently plaguing us, such as "How many MLR/PCA/PLS factors should I use in my model?", "How many degrees of freedom should I use in calculating an error (e.g., SEE, SEP, etc.)?", and "Can I transfer my calibration model?" (or more importantly and fundamentally: "How can I *tell* if I can transfer my calibration model?"), may never be answered in a completely rigorous and satisfactory fashion, but certainly improvements in the current state of knowledge should be attainable, with attendant improvements in the answers to such questions. New questions may arise which only fundamental statistical/probabilistic considerations may answer; one that has recently come to our attention is: "What is the best way to create a qualitative (i.e., identification) model, if there may be errors in the classifications of the samples used for training the algorithm?"

Part of the problem, of course, is that the statistical questions involved are very difficult and have not yet been solved completely and rigorously even by statisticians. Another part of the problem is that very few first-class statisticians are interested in, or perhaps even aware of, the existence of our subdiscipline or its problems. Thus, of necessity we push on and muddle through in the face of not always having a completely firm and mathematically rigorous foundation on which to base our use of the techniques we deal with. (Here comes our disclaimer): So we use these techniques anyway because

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This is for univariate data; what happens in the case of multivariate (multiwavelength) spectroscopic analysis? The same thing, only worse. To calculate the effects rigorously and quantitatively is an extremely difficult exercise for the multivariate case, because not only are the errors themselves are involved, but in addition the correlation structure of the data exacerbates the effects. Qualitatively we can note that, just as in the univariate case, the presence of error in the absorbance data will "bias the coefficient(s) toward zero," to use the formal statistical description. In the multivariate case, however, each coefficient will be biased by different amounts, reflecting the different amounts of noise (or, more generally, the error) affecting the data at different wavelengths. As mentioned above, these effects will be exacerbated by intercorrelation between the data at different wavelengths. The difficulty comes when you realize that it is not simply the correlations between pairs of wavelengths that are operative in this regard but also the intercorrelation effects of the data when the wavelengths are taken 3, 4, …, n at a time. This makes the problem intractable.

Now, we're sure that there are some readers who will read this and say something along the lines of "well, all you need do is do a PCA/PLS analysis and get rid of all those effects." Actually, there might be a germ of truth to that—if you can always do all your calibration modeling using only the first two or three PCA or PLS factors. Beyond that, you'll run into what we might informally call "the Law of Conservation of Error" (except for the fact that, as we all know, error is *much* easier to create than destroy!). In special cases, however, such as PCA and PLS, the total error really is constant, so that we quickly get into territory where the noise that you pushed out of the first couple of factors reappears and affects the higher factors even more than the original noise affected the original data.

So in the long-gone days of our academic lives, the chemical measurements, being based on high-accuracy gravimetric and volumetric techniques, were indeed the proper ones to put on the X-axis. Contrast that with the current state of technology: instruments have improved enormously, and rather than making up training samples by simple gravimetric dilutions, we often obtain our training, or reference, values through complicated analytical methodologies, which are themselves fraught with so much error that even in favorable cases, the error can be 5%–10% of the analytical value. In current practice, therefore, where the instruments have improved but the reference values haven't, the error in the reference lab values really is greater than the error in the absorbance data. For this reason it is now appropriate to reverse the positions of the concentration and absorbance values relative to their place in the calculation schema.

So it's the changing nature of the world and the types of analyses we do that dictate how we go about organizing the calculations we use to do them. This comes from fundamental considerations of the behavior of the modeling process, which the science of statistics can tell us about.

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CHAPTER

27 CHALLENGES: UNSOLVED PROBLEMS IN CHEMOMETRICS☆

We called the issues we plan to discuss in this chapter "unsolved" problems, but that may be incorrect. It may be, perhaps, more accurate to call them "Unaddressed Problems in Chemometrics." Calling them "unsolved" implies that attempts have been made to solve them, but those attempts were unsuccessful, possibly because these problems are too difficult or possibly because maybe we're not smart enough. Calling them "unaddressed," on the other hand, really gets to the heart of the matter: A number of issues have come to our attention that nobody seems to be paying any heed to. It may very well turn out that some of these problems are too difficult to solve at the current state of the art in chemometrics, and maybe we're really not smart enough, but at this point we don't know, and we'll never know if nobody tries. Our attention was drawn to these problems via various routes. Some arose from our own work on various projects. Some arose from discussions in any of the various online discussion groups. Some have been floating around in the backs of our minds for what seems like forever but only recently crystallized into something concrete enough to write down in a coherent manner so that it could be explained to somebody else. Answers: We have none, only questions. We bring up these points to stir up some discussion, and maybe even a little controversy, and certainly with the hope that we can prod some of our compatriots "out there" to tackle some of these.

Conspicuous by its absence is the question of calibration transfer, even though we consider it unsolved in the general sense, in that there is no single "recipe" or algorithm that is pretty much guaranteed to work in all (or even a majority) of cases. Nevertheless, not only are many people working on the problem (so that it is hardly "unaddressed"), but there have been many specific solutions developed over the years, albeit for particular calibration models on particular instruments and for particular analyses. Also, we do include some discussion of the problem in Chapters 101–106 in this book, so we don't need to beat up on this one here and now.

SO WHAT ARE THESE PROBLEMS?

1. The first one we mention is the question of the validity of a test set. We all know and agree (at least, we hope that we all do) that the best way to test a calibration model, whether it is a quantitative or a qualitative model, is to have some samples in reserve, that is, not included among.the ones on which the calibration calculations are based, and use those samples as "validation samples" (sometimes called "test samples" or "prediction samples" or "known samples"). The question is:

☆Adapted from Spectroscopy 13(10) (1998) 11–15.

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